
The Intervention Gap in Latent World Models

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Abstract

Planning-time intervention fidelity is a distinct, measurable property of a learned world model: whether the model’s own open-loop transitions move task variables the way matched environment interventions do. In the settings we test, it is neither revealed by reward fit nor ensured by task-anchored training. Across released TD-MPC2 checkpoint sizes, episode return falls as an operator-error diagnostic on task observables grows, while reward-prediction error stays small and nearly flat, and a self-supervised world model trained without task signal preserves the same operator substantially better than a task-anchored model on the shared task. A capture-gated matched-intervention audit then localizes what fails. On Cheetah, three LeWorldModel checkpoints capture the current task query and support decodable real intervention effects; however, their imagined five-step effects are worse than predicting no effect and worse than an environment-endpoint oracle. The failure is task-direction rotation with excess gain, not feature collapse. This severe pattern is conditional: five PreJEPA seeds retain an oracle-relative deficit without it, Finger Spin experiments extend the deficit beyond locomotion with heterogeneous severity across seeds, and shared-bank effect geometry is both candidate- and support-dependent. We also test practice-side questions. In DreamerV3 the posterior distribution, not its sample, carries the current query; ensemble disagreement ranks error only near training support; and a frozen support-aware score degrades held-out error ranking in both tested transfer directions while native disagreement remains informative in both. We conclude that intervention fidelity must be audited directly, capture-first, on the model’s native interface.

1 Introduction

A world model used for planning must do two different things: represent the current variables that matter to the task, and predict how candidate actions change them (Sutton, 1991; Ha and Schmidhuber, 2018). Standard training signals certify neither requirement separately. One-step prediction quality correlates poorly with control performance (Lambert et al., 2020), and a model can pass next-step diagnostics while its implicit transition structure is incoherent (Vafa et al., 2024). A task variable can be readable from a real encoded state even when the model assigns the wrong consequence to an intervention; conversely, a transition score is difficult to interpret when the relevant quantity was never recoverable from the model state in the first place.

This paper develops one claim across five sets of results. *Planning-time intervention fidelity, the property that a model’s own open-loop transitions move task variables the way matched environment interventions do, is distinct and measurable. Value-based training signals neither reveal nor ensure it; it can fail severely while reward fit, current-state capture, and overall feature alignment all look fine; and it therefore has to be audited directly, capture-first, on the model’s native interface.* Every empirical statement below is relative to a declared task query, physical horizon, intervention support, and native model interface; none is a population-of-architectures claim.

The argument proceeds in three steps. First, value-based signals neither measure intervention fidelity nor produce it. On a released TD-MPC2 checkpoint-size sweep, episode return falls as an operator-error diagnostic on task observables grows, while reward-prediction error stays small and nearly flat, and the operator diagnostic ranks checkpoints differently from Bellman-residual and value-slice metrics that agree with each other (Section 3). Nor does training against task signal secure the property: on a shared task, a self-supervised world model trained with no task signal preserves the same operator substantially better than a task-anchored model. The value channel neither detects the failure nor, on this comparison, prevents it.

Second, a capture-gated matched-intervention audit localizes what fails (Sections 2, 4, and 5). On Cheetah, three LeWorldModel checkpoints capture the current task query and support decodable real intervention effects, while their own imagined five-step effects are worse than predicting no effect at all and worse than an environment-endpoint oracle. The failure is not wholesale feature collapse: the complete predicted feature change remains strongly aligned with the real one while its task component is rotated to near-orthogonal and amplified. The severe pattern is conditional rather than universal. Five independently trained PreJEPA predictors retain an oracle-relative deficit without reproducing the severe conjunction; Finger Spin extends the oracle-relative gap beyond locomotion with heterogeneous severity across seeds; and on a shared evaluation, action-effect geometry is candidate- and support-dependent rather than an architecture ranking.

Third, the same program yields practical guidance for auditing and trusting such models (Section 6). In a DreamerV3 study, the categorical posterior distribution is the sole tested surface that supports the current query across checkpoints; its sample, its mode, and the deterministic state all fail. Ensemble disagreement ranks error near training support and deteriorates away from it, and a frozen support-aware augmentation of disagreement degrades error ranking on held-out families in both tested transfer directions, while architecture-native disagreement remains informative in both.

The contributions, in the order the argument uses them, are: (i) released-checkpoint evidence that reward fit can be silent about, and task-anchored training insufficient for, operator fidelity on task observables; (ii) a capture-gated matched-intervention audit that separates target availability, real-effect resolvability, and fidelity of the model’s own transitions; (iii) a prospective Cheetah/LeWorldModel dissociation of current-query capture from severe propagation distortion, characterized as task-direction rotation with excess gain; (iv) boundary results (a PreJEPA non-replication of the severe conjunction, heterogeneous Finger severity, and support-dependent shared-bank geometry) that scope the phenomenon without erasing its oracle-relative deficit; and (v) interface and uncertainty results, including a bidirectional held-out falsifier for a frozen support-aware score, that turn the audit into usable practice.

2 Two Audits of Planning-Time Fidelity

Let x denote a Markov state or a declared sufficient observation history. Fix a family of task queries $\mathcal{Q} = \{q_1, \dots, q_m\}$, a physical horizon H , and a distribution ν over roots and action sequences. For an action a , define the environment operator

$$(\mathcal{K}_a q)(x) = \mathbb{E}[q(X_{t+1}) \mid X_t = x, \text{do}(A_t = a)], \quad (1)$$

and let $\hat{\mathcal{K}}_M$ denote the corresponding operator induced by candidate M ’s own open-loop transition on its planner-exposed state S_M . Every diagnostic statement in this paper is relative to $(\mathcal{Q}, H, \nu)$ and to the model’s native interface. Open-loop latent rollouts are evaluated because they are the object planners and imagined-rollout learners consume (Hafner et al., 2019, 2020; Hansen et al., 2022).

We use two instruments of increasing resolution.

2.1 Operator error on task observables

The coarser instrument asks how far $\hat{\mathcal{K}}_M$ is from $\mathcal{K}$ on task observables, without asking why. Ridge probes with grouped cross-validation decode each query from the model’s latent state; the model then advances its latent through its own native prediction mode to the planning-time horizon, and the probe-decoded outcome is compared with the same query evaluated on the real environment outcome under the same actions. The result is a single operator-error score per candidate on the declared query family. Probe quality is reported separately from operator error, so a poorly decodable observable is

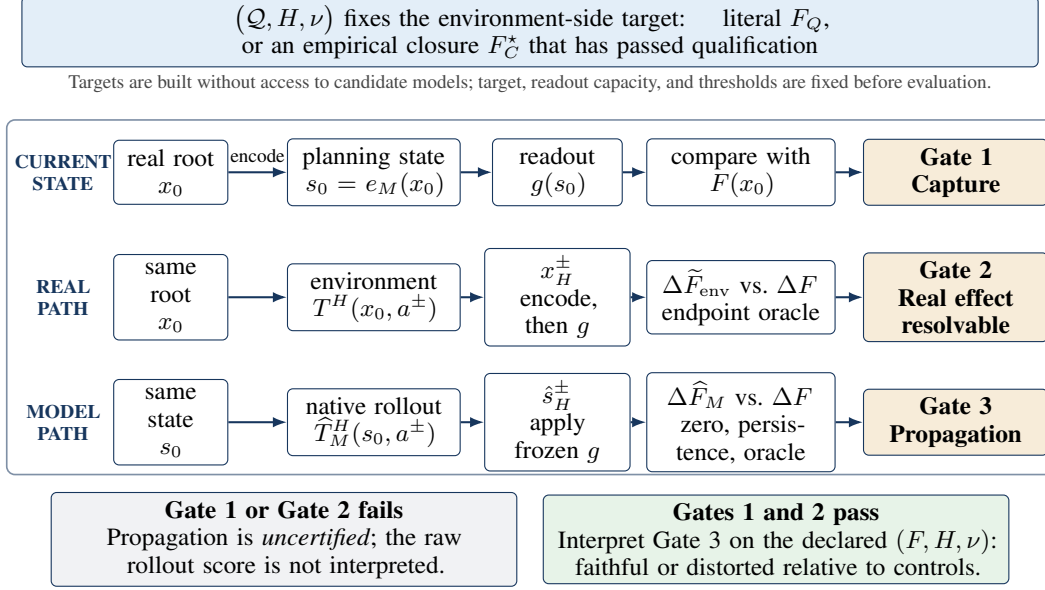


Figure 1: **Capture-gated matched-intervention audit.** A query family, physical horizon, and intervention support define an environment-side target before candidate evaluation. Gate 1 tests whether the target is readable from held-out real planning states. Gate 2 tests whether the frozen endpoint readout resolves the real effect of the matched actions. Only when both prerequisites pass is Gate 3 interpreted: the model’s native imagined effect is compared with the real effect, zero effect or persistence, and the environment-endpoint oracle. Failure of either prerequisite yields uncertified propagation rather than evidence of faithful propagation.

not silently charged to the transition. This diagnostic is deliberately complementary to value-based metrics: it shares their task-relativity but scores the transition on task observables rather than on reward, value, or return (Farahmand et al., 2017; Grimm et al., 2020).

The coarse score conflates two failure sources: the information may be absent from the latent state, or present but moved wrongly by the transition. Section 3 uses the coarse score, where that conflation does not matter for the conclusions drawn; the rest of the paper uses the finer instrument, which separates the sources.

2.2 The capture-gated matched-intervention audit

The finer instrument decomposes fidelity into three separately testable prerequisites, summarized in Figure 1.

Capture. A prespecified readout is trained on real encoded states and evaluated on held-out roots. Capture reports coordinatewise and aggregate error under development-only normalization.

Real-effect resolvability. Before model propagation is interpreted, the matched environment intervention must produce an effect that the frozen endpoint readout can resolve at real environment endpoints. If this check fails, the corresponding propagation comparison is not interpreted.

Propagation. From the same real root, an action sequence is executed in the environment and through the model’s own open-loop transition. The frozen readout maps the imagined endpoint to the target coordinates. Model error is compared with persistence, with zero effect, and with the error of the real environment endpoint under the same readout (the environment-endpoint oracle).

The gate order follows the structure of finite-horizon query error, which obeys the schematic decomposition

$$\|\hat{q}_h - q_h\| \leq \|B_h\| P_{0:h} \epsilon_{\text{cap},0} + \sum_{t=0}^{h-1} \|B_h\| P_{t+1:h} (\epsilon_{\text{op},t} + \tau_{C,t}) + \tau_{q,h}, \quad (2)$$

where $\epsilon_{\text{cap},0}$ is capture error at the real root, $\epsilon_{\text{op},t}$ is closure-restricted transition error, τ -terms are approximation residuals, and $P_{s:h} = \prod_{j=s}^{h-1} L_j$ collects environment stability factors (Appendix A). A propagation score is interpretable only when the capture term is small; the decomposition identifies measured error sources and is not a guarantee about realized return.

Matched interventions and supports. Each environment family supplies repeated roots and action contrasts. The same action sequence, mapped into each model’s action interface, is applied to the environment and to every compatible candidate. Task-policy and environment-random supports remain separate groups; roots, actions, horizons, and model outputs are repeated measurements within a family. The action contrasts test action sensitivity in the spirit of controllability metrics for generative world models (Bruce et al., 2024; Arai et al., 2024), but against executed environment endpoints rather than alternative model rollouts.

Isolation, readouts, and inference. Closure development, closure validation, probe training, calibration, and evaluation use disjoint environment-family identities; candidate models are unavailable during environment-side target construction, and evaluation data affect no target rank, readout capacity, normalization, threshold, candidate inclusion, or action support. The primary readout is a standardized ridge probe with grouped nested cross-validation; controls include label and feature shuffles, horizon-zero identity, persistence, executed-environment endpoints, wrong-action and action-null contrasts, probe-capacity sensitivity, and direct predicted-versus-real feature alignment. The complete environment family is the uncertainty unit; checkpoints, roots, actions, coordinates, and imagined samples are never treated as independent replications, and each candidate is reported separately.

Calibration and closure boundary. The audit is calibrated on synthetic systems before application to learned models: ten prespecified acceptance cases require it to distinguish capture failure from propagation failure, reject persistence as successful dynamics, recover a known stable rank, remain invariant to orthogonal basis changes, and halt on split leakage; all ten pass under the fixed configuration (Appendix B). Learned-model results use the literal queries F_Q . A larger environment-side closure target $\mathcal{C}_{Q,H,\nu} = \text{span}\{\mathcal{K}_{a_0} \cdots \mathcal{K}_{a_{t-1}} q\}$ is used only after passing rank, null, and stability checks; the original Cheetah construction did not pass those checks, and the qualified successor holds only on matched action shells (Appendix F).

3 The Value Channel Neither Reveals Nor Ensures Operator Fidelity

This section uses the coarse operator diagnostic of Section 2.1 to establish the limitation that motivates the finer audit. Both results are single-environment observations on Cheetah locomotion; they flag a failure mode rather than prove a general law.

3.1 Return falls with operator error where reward fit is silent

The released multi-task TD-MPC2 checkpoints (Hansen et al., 2024) provide five model sizes trained under one recipe. Across this sweep, episode return on cheetah-run falls as operator error on task observables at the planning-time horizon grows, with rank correlation -0.90 , and the association survives controlling for reward-prediction accuracy. Reward-prediction error itself is small and nearly flat across the same checkpoints (Figure 2). The largest checkpoint is the sharpest instance: its reward fit is among the best in the sweep while its operator error is the worst and its planning return collapses. On this sweep, the value channel’s own fit statistic is silent exactly where the transition moves task observables wrongly.

The operator diagnostic also measures something the standard model-quality proxies do not. Ranking the same checkpoints by operator error on the full observable family agrees only weakly with a Bellman-residual ranking, while a value-restricted variant of the operator score reproduces the

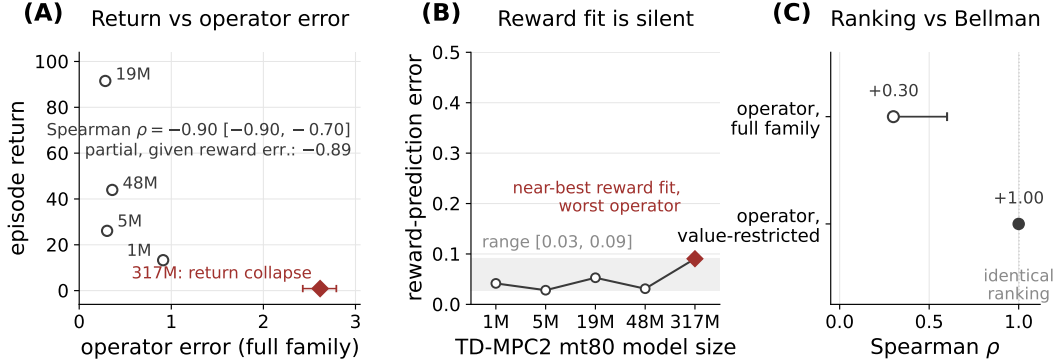


Figure 2: **Reward fit is silent where the operator fails.** Released multi-task TD-MPC2 checkpoints at five sizes on cheetah-run (dependent sizes; descriptive sweep). (A) Episode return against operator error on task observables at the planning-time horizon; rank correlation -0.90 (anchor-bootstrap 95% interval $[-0.90, -0.70]$; leave-one-out range ≥ -0.80); the association survives controlling for reward-prediction accuracy (partial rank correlation -0.89). The 317M checkpoint is marked: worst operator error, collapsed return. (B) Reward-prediction error across the same checkpoints, flat within $[0.03, 0.09]$, with the same checkpoint marked: near-best reward fit, worst operator. (C) Checkpoint rankings: full-observable operator error versus a Bellman-residual metric (rank correlation $+0.30$, 95% interval $[+0.30, +0.60]$), and a value-restricted operator variant versus the same Bellman residual ($+1.00$). One environment and $n = 5$ dependent sizes; correlational evidence only.

Bellman-residual ranking almost exactly (Figure 2, right panel). This disagreement is itself informative: restricted to value-relevant directions, the operator score reproduces the value-channel ranking, so its additional signal comes from the task-observable directions the value channel does not constrain. We therefore treat the operator score as a complement to value-based metrics, to be reported alongside them, not as a replacement (Farahmand et al., 2017; Grimm et al., 2020; Lambert et al., 2020).

These are five dependent released sizes from one training recipe on one task; the sweep supports a correlational flag, not a selection rule or a return guarantee.

3.2 Task-anchored training did not confer operator fidelity

Reward fit does not reveal operator error; a remaining possibility is that task-anchored training nevertheless prevents it. On this comparison it did not. Three independently seeded LeWorldModel runs (Maes et al., 2026), self-supervised video-prediction world models trained from scratch on cheetah-run with no task signal, are compared with a task-anchored TD-MPC2 model trained on the same task, using matched ridge probes on shared task observables and each model’s native prediction mode at the planning-time horizon. The self-supervised models preserve the operator substantially better: their operator error is roughly 0.46 of the task-anchored model’s, with disjoint 95% intervals across all three seeds (Figure 3).

Three control analyses support the comparison. A single-task and a multi-task anchored checkpoint give the same answer, ruling out a multi-task handicap. Restricting the observable family to coordinates both models probe well makes the gap slightly larger, not smaller, ruling out probe asymmetry. The seed-level intervals remain disjoint under a one-hidden-layer MLP probe in place of the ridge probe. The comparison still spans only two model families with one anchored seed, on models with different input modalities and native prediction modes; it bounds the claim “task anchoring secures operator fidelity,” and supports nothing stronger.

3.3 Limits of the coarse diagnostic

Together the two results say the value channel neither detects operator error nor, on this comparison, prevents it. But the coarse score cannot say what fails: an operator-error number conflates information that is absent from the latent state with information that is present and moved wrongly, and it

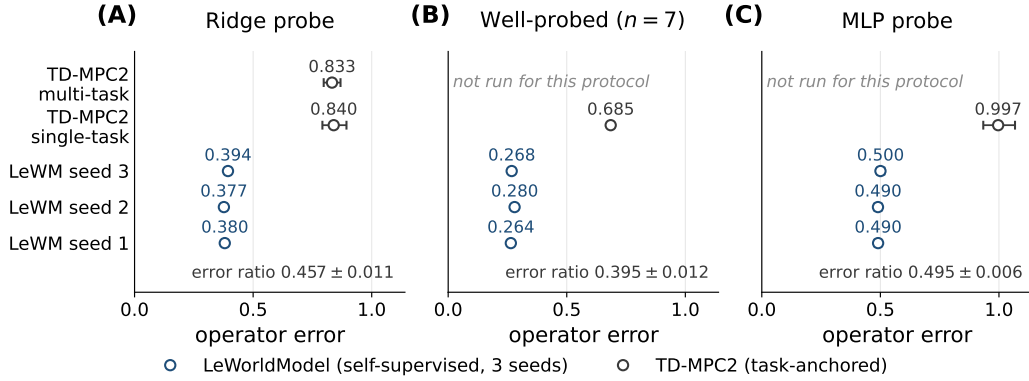


Figure 3: Task anchoring did not confer operator fidelity. Operator error on shared task observables at the planning-time horizon; blue marks the self-supervised LeWorldModel seeds throughout. (A) Ridge probes on the full shared family: three independently seeded LeWorldModel runs (0.384 ± 0.009) versus task-anchored TD-MPC2 (single-task 0.840; multi-task 0.833); error ratio 0.457 ± 0.011 . (B) The family restricted to the $n = 7$ observables both models probe well, against the single-task anchored checkpoint (the multi-task variant was not run under this protocol): the gap grows (ratio 0.395 ± 0.012). (C) One-hidden-layer MLP-probe ablation, same comparator: seed-level 95% intervals remain disjoint (ratio 0.495 ± 0.006). Two model families, one anchored seed per variant, differing input modalities and native prediction modes; a bound on the anchoring claim, not an architecture ranking.

Table 1: Capture-gated matched-intervention outcomes. Each row is one task–architecture roster. Candidates are not pooled, and raw scores are not comparable across tasks. Intervals use 4,096 whole-family bootstrap resamples on $n = 96$ evaluation families per candidate. Severe means predicted-effect R^2 and its 95% upper endpoint are both below zero.

Task / model	Capture R^2	Real R^2	Predicted R^2	Severe	Oracle deficit
Cheetah / LeWM	0.579–0.609	0.127–0.198	−6.33 to −3.36	3/3	3/3
Cheetah / PreJEPA	pass	≈ 0.390	−0.005 to 0.064	0/5	5/5
Finger / LeWM	0.539–0.555	0.733–0.789	−0.869, 0.287, 0.367	1/3	3/3

aggregates over a support without matching interventions. The remainder of the paper applies the capture-gated audit of Section 2.2, which separates those readings. The separation matters concretely: the self-supervised family that preserves the coarse operator better in this section is the same family whose matched intervention effects fail most severely in the next. The two instruments measure different properties of the same models; that is the reason to keep both, and the reason neither raw score is comparable across instruments.

4 Capture Without Propagation

The capture-gated audit localizes the failure that the coarse diagnostic cannot separate: a model can represent the task information and still mispredict how actions change it. Both tasks in this and the next section come from the DeepMind Control Suite (Tassa et al., 2018); Table 1 summarizes the three prospective rosters.

4.1 The primary dissociation

Three frozen LeWorldModel checkpoints (Maes et al., 2026) are evaluated on Cheetah task-policy support. Each evaluation family supplies three roots and six matched actuator contrasts at a physical horizon of five environment steps; the frozen target is endpoint torso speed, and readouts and thresholds are fixed on 48 calibration families before each checkpoint is scored on 96 evaluation families.

All three checkpoints pass both prerequisites: the current query is readable from real encoded roots, and the frozen readout resolves the real effects of the matched pulses at real endpoints. Their own imagined effects then fail both confirmatory comparisons. Predicted-effect R^2 is far below zero, worse than predicting no effect at all, with every whole-family 95% interval below zero, and every checkpoint’s imagined endpoint is reliably worse than the environment-endpoint oracle under the same readout (Table 1; Figure 4, top row). Current-query capture and real-effect resolvability therefore coexist with severe distortion of the model’s own five-step action effects, prospectively and across the roster.

4.2 Rotation with excess gain, not collapse

The simplest interpretation, that the transition loses the intervention entirely, is not supported. In descriptive controls frozen with the confirmatory package, the complete predicted feature change remains strongly aligned with the real one (contrast cosine 0.85–0.89 across the three checkpoints), while the same change projected onto the task readout is nearly orthogonal to the real task effect and 1.7–2.6 times too large. Within the measured surface, the failure is task-direction misalignment with excess gain: the model moves its features broadly the right way and moves the task-relevant component of that motion in the wrong direction, too strongly. This is why a generic feature-space rollout error would have looked benign here, and why trajectory-plausibility scores can miss the failure entirely (Section 7). The characterization is descriptive; it constrains candidate mechanisms without establishing one.

What the primary result does not establish is bounded in Section 8: no absence of task information under every nonlinear decoder, no claim at other horizons or supports, no planner-failure or return claim, and no generalization beyond the tested roster, which is the subject of the next section.

5 The Deficit Replicates; Severe Distortion Does Not

The severe Cheetah pattern could have been a universal property of latent world models, a property of one architecture, or an artifact of one roster. Three boundary results place it: the oracle-relative deficit reappears in every tested roster, while the severe conjunction is conditional on model family, task, and seed.

5.1 A prospective non-replication in PreJEPA

Five independently initialized action-conditioned predictors over a shared frozen DINOv2 backbone (Oquab et al., 2024) are evaluated with the same four-part decision structure, with candidate selection completed before any matched-intervention outcome is seen. Every predictor passes root capture, real-effect resolution, and the oracle-relative comparison; none satisfies the severe below-zero condition observed in LeWorldModel (Table 1; Figure 4, bottom row). The severe conjunction therefore does not replicate in a materially different visual predictor, under a prespecified test capable of detecting it. Accurate propagation is not established either: every predicted-minus-oracle error interval is strictly positive, so all five imagined endpoints remain reliably worse than the real endpoints they try to anticipate. Where LeWorldModel rotates and amplifies the task effect, PreJEPA attenuates it toward zero: two different geometries behind one shared deficit.

5.2 A second task: heterogeneous severity on Finger Spin

The same capture-gated protocol applied to Finger Spin, with endpoint negative hinge velocity as the frozen query and three independently seeded LeWorldModel runs, reproduces the structure of the dissociation beyond locomotion. All three runs pass root capture and real-endpoint resolution, and every paired predicted-minus-oracle interval is strictly positive. Severity, however, is not seed-invariant: one run meets the severe below-zero certificate while the other two retain positive but oracle-inferior predicted-effect signal (Table 1; Figure 5). The oracle-relative gap replicates across tasks; the extreme form does not replicate even across seeds of one architecture on one task.

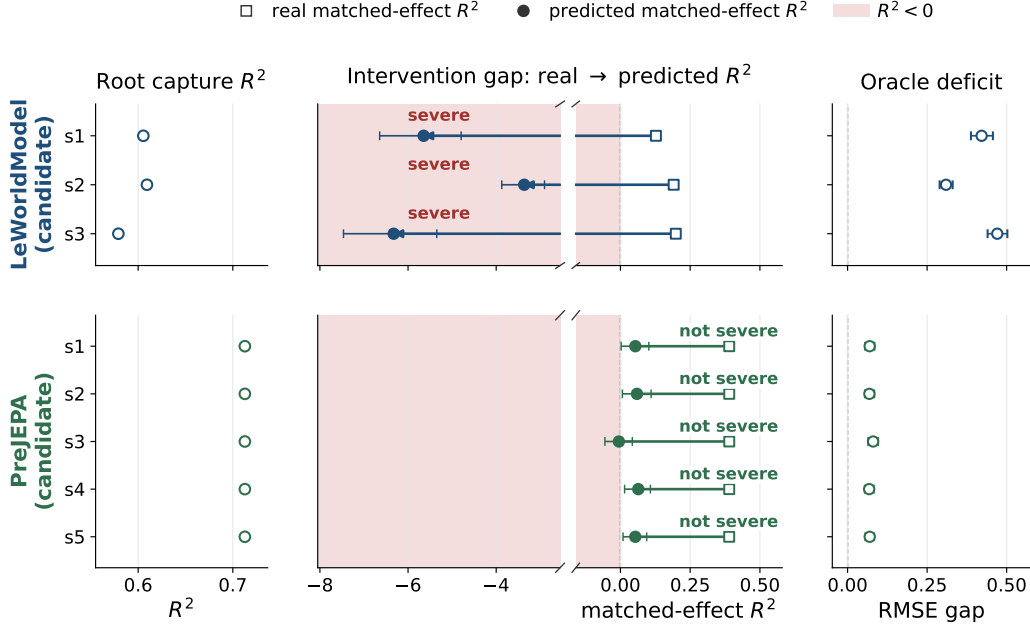


Figure 4: **Intervention gap and non-replication on Cheetah.** Rows are architectures. The left column is current-query capture R^2 . The middle column is the matched-effect gap: open squares are real matched-effect R^2 and filled circles are predicted matched-effect R^2 (family 95% intervals). Arrows run from the environment effect to the model’s predicted effect. That panel uses a broken R^2 axis so both ends remain readable; the omitted interval contains no plotted values, and the horizontal scale is shared across architectures within each pane. The shaded region is $R^2 < 0$. Tags mark the frozen below-zero distortion gate: severe means predicted R^2 and its 95% upper endpoint are both < 0 ; not severe means that conjunction does not hold. The right column is the oracle deficit (paired predicted-minus-oracle RMSE difference for LeWorldModel; predicted-minus-oracle RMSE for PreJEPA); both > 0 means worse than the environment-endpoint oracle. Candidate labels are not comparable across rows. Intervals use 4,096 whole-family bootstrap resamples ($n = 96$ evaluation families within each candidate). Readouts and thresholds were frozen on calibration before evaluation. LeWorldModel shows capture and real-effect resolution with predicted effects worse than zero and worse than the oracle. PreJEPA retains capture, real-effect resolution, and oracle deficit but does not reproduce the severe conjunction. Two architectures do not license a population-of-architectures claim; PreJEPA non-replication is not evidence of accurate propagation.

5.3 Shared-bank geometry is candidate- and support-dependent

The confirmatory rosters above use valid but different environment families. A shared evaluation places the frozen LeWorldModel and PreJEPA candidates, together with a descriptive panel of released TD-MPC2 models (Hansen et al., 2024), on identical roots, five-step actions, and environment endpoints, with task-policy and environment-random supports analyzed separately. No unanimous architecture signature emerges (Figure 6). LeWorldModel task-policy cells fail real-effect resolution on this bank and remain uncertified for propagation; PreJEPA task-policy effects are consistently attenuated; TD-MPC2 exhibits large checkpoint-to-checkpoint and support variation, including a sign-reversal and extreme-amplification pattern in its largest released checkpoint across supports, with no monotonic size pattern. Every prespecified architecture-by-support summary remains inconclusive.

Together, these boundary results support the following conclusion: capture and propagation remain separable wherever tested, and an oracle-relative propagation deficit has appeared in every qualified roster, but the severity and geometry of the failure are properties of the candidate, task, seed, and support, not of latent world models as a class. The sharp Cheetah classification is valid on its frozen support and is not a support-invariant property of the model.

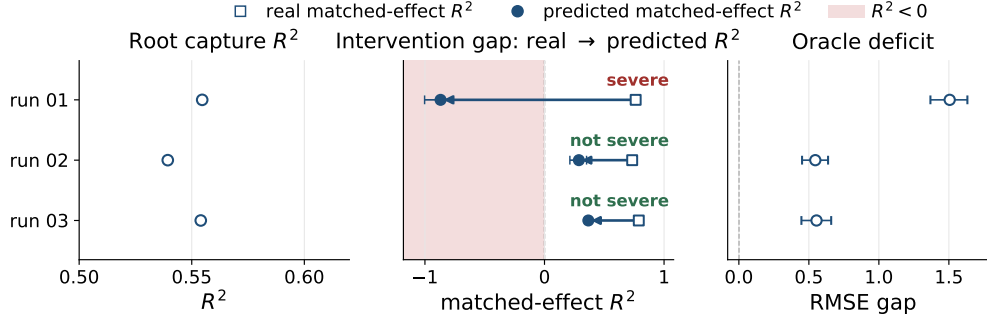


Figure 5: **Finger Spin: oracle-relative gap with heterogeneous severity.** Three independently seeded LeWorldModel runs on Finger Spin (endpoint negative hinge velocity). Columns as in Figure 4: capture R^2 ; matched-effect gap with the frozen severe/not-severe gate; paired predicted-minus-oracle RMSE difference (> 0 means worse than the environment-endpoint oracle). Intervals use 4,096 whole-family bootstrap resamples ($n = 96$ evaluation families per candidate); readouts and thresholds frozen on calibration before evaluation. Raw scores are not comparable to Cheetah. The oracle-relative deficit replicates in every run; severe distortion appears in one of three. One architecture and one additional task; no universal mechanism, architecture ranking, or planner claim.

6 Auditing in Practice: Surfaces, Uncertainty, and Support

If intervention fidelity is conditional and cannot be read off the value channel, the practical question is how to audit it and when to trust the model’s own signals. Three results answer parts of that question: which surface to read, how far disagreement can be trusted, and which gate binds first.

6.1 The query is carried by the posterior distribution, not its sample

Auditing presupposes a state surface that carries the query. In a DreamerV3 study (Hafner et al., 2023), six native surfaces were localized for the current query across four training checkpoints under frozen capture gates. Centered posterior logits are the sole surface passing both gates at every checkpoint; the sampled categorical latent, its mode, the deterministic recurrent state, and their concatenation pass at none, and finite sample budgets through 1,024 samples do not bridge to the exact-logit result (Figure 9; Appendix E). The information the audit needs lives in the posterior *distribution*, not in any single sampled state the rollout machinery consumes. This is a representation-boundary result: it licenses a distribution-level readout for auditing and does not test Dreamer propagation, which remains uncertified.

6.2 Disagreement ranks error only near support

Ensemble disagreement is the standard epistemic signal for learned dynamics (Lakshminarayanan et al., 2017; Chua et al., 2018). Within the five-seed PreJEPA family it behaves as a sharply support-relative diagnostic: disagreement ranks future-feature error under task-policy support ($\rho = 0.678$) and is inconclusive under environment-random support ($\rho = 0.027$); a follow-up grid that independently varies root-context and future-action departure shows the tracking deteriorating along both axes (Figure 7). Predicted branches also contract relative to encoded real branches and move toward the offline-training support while moving away from same-support environment trajectories; the prespecified joint attractor signature is not established. Within this family, a prespecified equal-weight combination of disagreement percentile and support score improves global error ranking over disagreement alone. None of this calibrates epistemic uncertainty; it bounds where the proxy is informative.

6.3 A frozen support correction fails held-out transfer in both directions

The within-family success suggests a general rule: add a support term to disagreement everywhere. A bidirectional held-out test falsifies that frozen rule. The equal-weight score is frozen on PreJEPA and transferred without refitting, first to three independently trained held-out TD-MPC2 runs, then

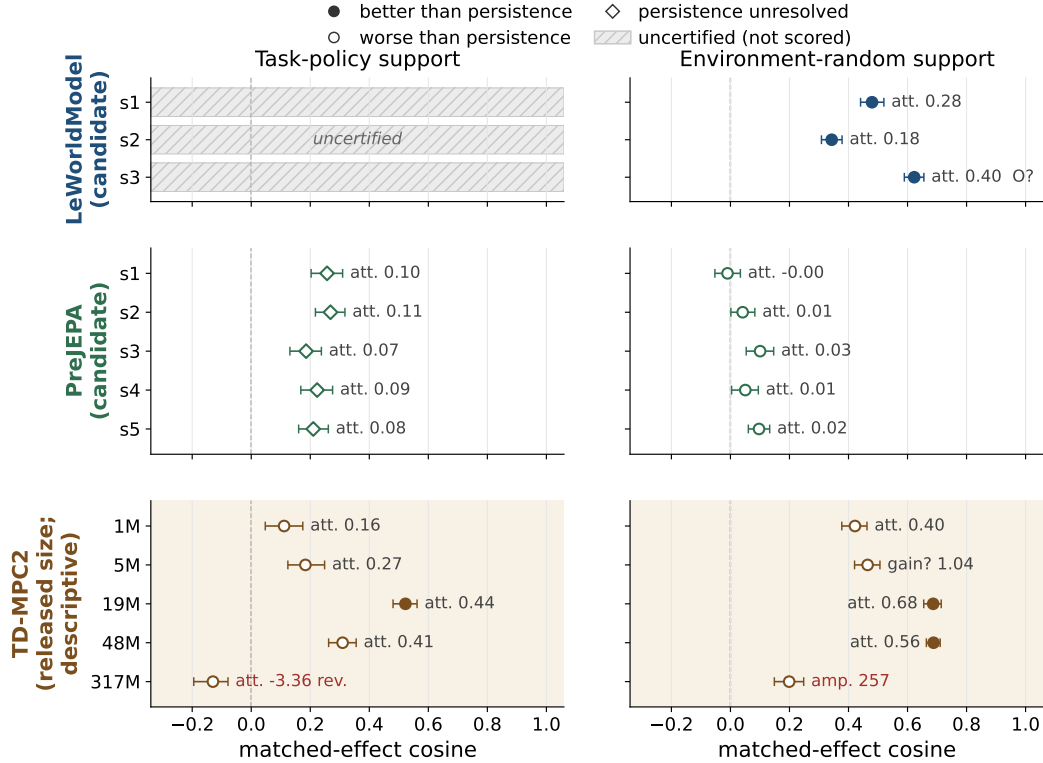


Figure 6: **Shared-evaluation action-effect geometry.** Rows are architectures and columns are intervention supports on identical roots, five-step actions, and environment endpoints. Within each column the horizontal axis is matched-effect cosine and is shared across architectures; the vertical axis lists candidates and is not comparable across rows (three LeWorldModel checkpoints, five PreJEPA seeds, five dependent TD-MPC2 released sizes). Intervals use 4,096 whole-family bootstrap resamples on the complete reset family (256 permutations). Hatched cells fail real-effect resolvability or capture on this bank and are uncertified for propagation; they are not scored as propagation successes or failures. Numeric tags are frozen gain (att. attenuation, amp. amplification); “rev.” marks a negative-direction label; “O?” marks an unresolved oracle comparison. Marker fill encodes the persistence comparison. Thirteen primary-core cells meet inclusion criteria; every architecture-by-support summary remains inconclusive. The TD-MPC2 row is descriptive only (dependent released sizes). This figure does not rank architectures or establish a size law.

reciprocally, with TD-MPC2 joining development, to three independently seeded held-out LeWorld-Model runs on a fresh environment bank. Its ingredients are the standard off-support penalties: native disagreement (Yu et al., 2020; Kidambi et al., 2020) and a feature-space support distance descended from Mahalanobis out-of-distribution scores (Lee et al., 2018).

In both directions the same pattern holds: native disagreement ranks endpoint error informatively, the frozen combined score is modestly but reliably worse than disagreement alone, and support distance alone is much weaker (Figure 8; Appendix C). In the reciprocal direction every individual held-out run has a strictly negative combined-minus-native interval. The two directions share one task and bank design, so they bound this frozen additive rule rather than every support correction; they do not establish a universal no-support law. The practice they support is conservative: treat architecture-native disagreement as the default transferable error-ranking signal, and treat any support augmentation as a family-specific tool whose transfer must be revalidated before reuse.

6.4 Capture is the first gate to fail in practice

The gate order of Section 2.2 is also where audits fail in practice. In a matched-shell audit of three independently trained TD-MPC2 runs, every primary cell terminated at the first gates: absolute root

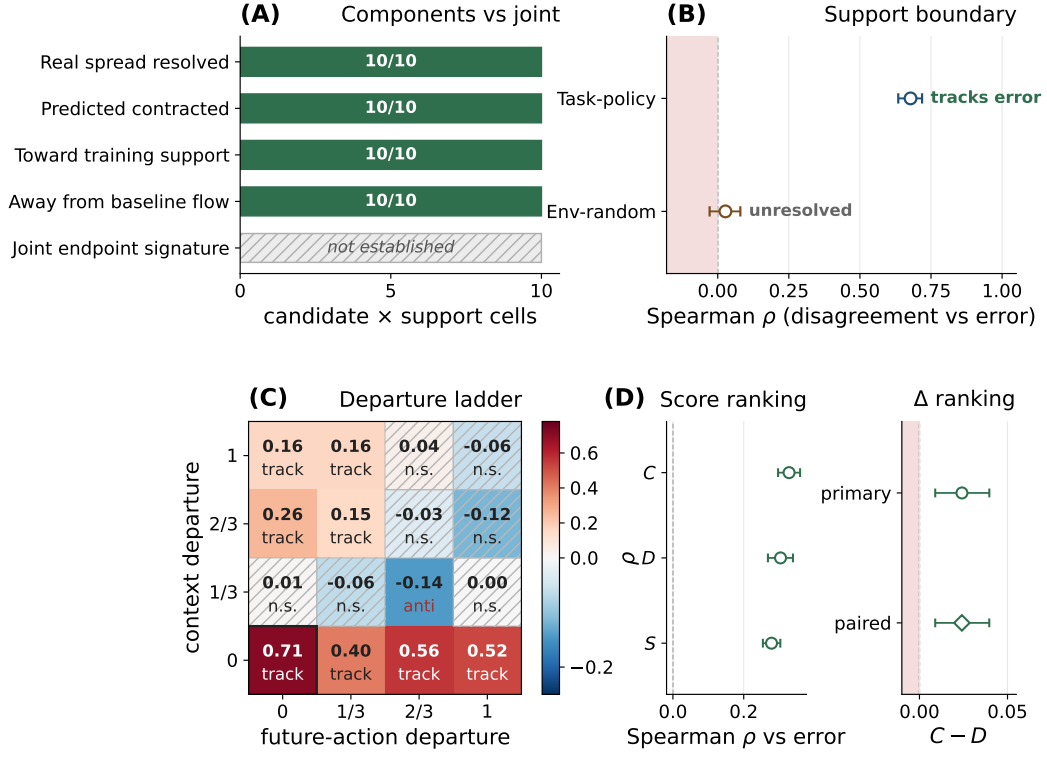


Figure 7: Support-linked signatures and departure ladder. (A–B) PreJEPA endpoint/support analysis (five candidates; complete environment family; 4,096 whole-family bootstraps). All ten candidate–support cells resolve real branch spread, contract predicted spread, move toward training support, and move away from same-support baseline flow; the prespecified joint endpoint signature is not established. Panel B is family-level Spearman ρ between seed disagreement and future-feature error under intervention (task-policy $\rho = 0.678$, [0.635, 0.719]; environment-random $\rho = 0.027$, [−0.029, 0.080]). (C–D) Context–action support ladder ($n = 384$ evaluation families; 4×4 grid; 96 families per cell; primary render lane). Cell color is Spearman ρ ; hatched cells are unresolved; the boxed corner is near-policy support. Context and future-action components are each detectable while their interaction is inconclusive. $C = 0.5D_{\text{pct}} + 0.5S$ improves global ranking over disagreement D ($\Delta = 0.0241$, 95% interval [0.0090, 0.0397]); the paired lane is a technical sensitivity, not an independent replication. This is within-PreJEPA evidence only and does not calibrate epistemic uncertainty.

capture passed on task-policy roots but real-endpoint effect resolution failed at both tested action radii, and on environment-random roots absolute capture itself failed, so no propagation cell was interpretable in either direction (Appendix F). A readout qualified for broad state reconstruction on one bank did not guarantee either support transfer for the scalar query or sufficient resolution after endpoint-minus-root differencing. The qualified environment-side closure result shows the same limitation at the target level: closure held on matched action shells, not on the full action ball.

The resulting protocol is brief. Declare the query, physical horizon, and support before any candidate is evaluated. Qualify the readout twice, on real roots and on real observed endpoints, per support. Interpret imagined propagation only for cells that pass both, against persistence, zero effect, and the environment-endpoint oracle. Read the model’s native interface and, where the state is distributional, the distribution rather than its sample. Treat disagreement as informative only near training support, and require any support correction to qualify on held-out families before use. Fidelity, throughout, is a relation among model, query, horizon, support, and interface, not a single score.

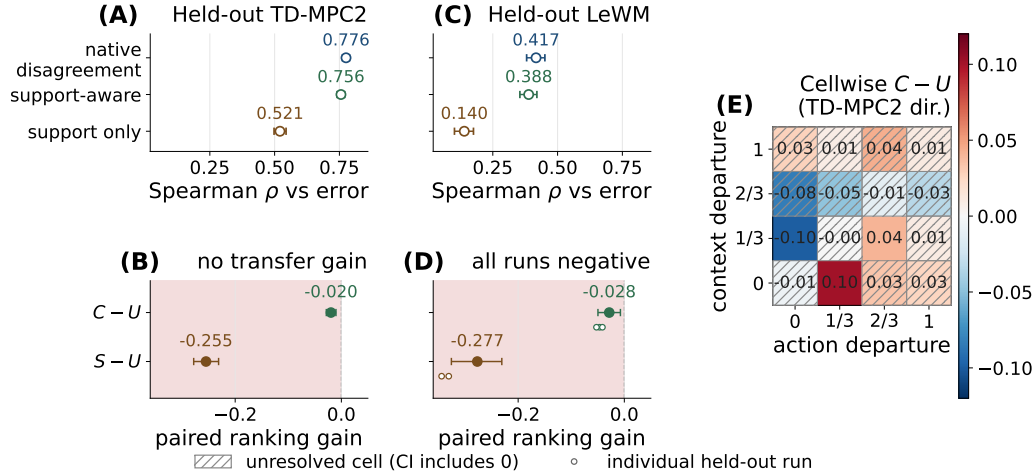


Figure 8: **Frozen support augmentation fails held-out transfer in both directions.** The score $C = 0.5U_{\text{pct}} + 0.5S$ is frozen on PreJEPA and applied without refitting. (A–B) Held-out TD-MPC2 (three independently trained 4M runs). Complete-role Spearman ρ between each score and endpoint error: native disagreement U 0.776 ([0.762, 0.789]); frozen combined C 0.756 ([0.741, 0.770]); support distance alone 0.521 ([0.498, 0.544]). Paired ranking gain relative to U : $C - U = -0.020$ ([−0.028, −0.010]); the shaded region is $\Delta \leq 0$. (C–D) Reciprocal held-out LeWorldModel (three independently seeded epoch-100 runs on a fresh environment bank). The pattern repeats at a lower ranking level: U 0.417 ([0.381, 0.451]); C 0.388 ([0.355, 0.420]); support alone 0.140; gain -0.028 ([−0.050, −0.007]). Open circles are individual held-out runs; each has a strictly negative combined-minus-native interval (Appendix C). (E) Context $\times$ action map of $C - U$ in the TD-MPC2 direction. Hatched cells have a 95% interval that includes zero and are unresolved; resolved cells remain mixed in sign and do not reverse the complete-role label. The ρ axes in A and C are shared so the lower reciprocal level is visible; the two directions are analyzed separately and never pooled. Intervals use 4,096 whole-family context-stratified bootstrap draws ($n = 384$ evaluation families per direction; 4,608 context–action rows as repeated measures). One task and bank design per direction; not calibration, not a universal no-support law, and silent on reward, return, and decision value.

7 Related Work

World models for planning. Learning a model of action consequences and planning inside it is a long-standing program (Sutton, 1991; Ha and Schmidhuber, 2018). Modern latent world models couple an encoder with an open-loop latent transition that a planner or imagined-rollout learner consumes directly (Hafner et al., 2019, 2020, 2021, 2023; Schrittwieser et al., 2020; Hansen et al., 2022, 2024). The audit evaluates exactly this planner-exposed interface: real encoded roots and the model’s own open-loop transitions.

Value equivalence and objective mismatch. Value-aware and value-equivalent model learning restrict model accuracy to the policy and value classes relevant for planning (Farahmand et al., 2017; Grimm et al., 2020, 2021; Silver et al., 2017), and one-step likelihood is known to correlate poorly with control performance (Lambert et al., 2020). Our distinction is operational: the evaluation target is constructed outside candidate models, representational capture is tested on real planning states, and transition propagation is interpreted only after capture and persistence checks. The contribution is not the existence of function-relative operator error, but a prospective measurement protocol that separates target availability, real-effect resolution, and fidelity of the model’s own transitions. Section 3 adds a released-checkpoint instance of the mismatch: reward fit can be small and nearly flat across checkpoints whose operator error and planning return vary widely.

Rollout error and model exploitation. One-step-accurate learned models can compound error under the distribution the planner induces (Ross and Bagnell, 2012; Janner et al., 2019), produce

unstable multi-step sample rollouts (Talvitie, 2014, 2017), and be exploited by policy optimization in regions where they are wrong (Kurutach et al., 2018). These accounts concern training and control performance in aggregate; the present protocol instead measures the propagation side directly, per query and per matched intervention, and only after capture and real-effect prerequisites pass.

Evaluating world models beyond return. A growing line evaluates world models against structural criteria rather than next-step loss or realized return. Vafa et al. (2024) show that generative sequence models can pass next-step diagnostics while their implicit transition structure is incoherent, in a discrete-automaton setting; the present audit is a continuous-control, intervention-matched analogue with explicit capture gating and persistence and oracle comparators. Yu et al. (2026) argue, as a position, that world-model evaluation should center interventional reasoning and rollout validity; we contribute an executed, prospectively gated instance of such an audit together with its boundary conditions. Zhang (2026) probe the latents of learned environment simulators and intervene on latent directions; our interventions are instead matched in the environment and scored at its executed endpoints. Action-sensitivity metrics for generative interactive models (Bruce et al., 2024; Arai et al., 2024) share the concern that learned models under-use action inputs; the matched-contrast design tests this against real action consequences rather than against alternative model rollouts.

Operator-theoretic language. Koopman and dynamic-mode methods lift dynamics to linear operators on finite observable families and estimate them from data (Koopman, 1931; Schmid, 2010; Williams et al., 2015; Mezić, 2005; Korda and Mezić, 2018; Brunton et al., 2022). We use that language to estimate a finite-horizon environment-side closure, not to claim global linearity or spectral recovery.

Probing internal representations. Linear-probe methodology motivates explicit controls for readout capacity and selectivity (Alain and Bengio, 2017; Hewitt and Liang, 2019; Belinkov, 2022). Probes have recovered linearly readable internal world state in sequence models (Li et al., 2023; Nanda et al., 2023), while amnesic probing shows that readability does not imply behavioral use (Elazar et al., 2021). The capture gate is therefore paired with a separate propagation test rather than read as evidence that decoded information is used.

Uncertainty and support. Ensemble disagreement is the standard epistemic signal for learned dynamics (Lakshminarayanan et al., 2017; Chua et al., 2018) and is used to penalize or truncate off-support rollouts in offline model-based reinforcement learning (Yu et al., 2020; Kidambi et al., 2020). Its reliability degrades under distribution shift (Ovadia et al., 2019), and implemented disagreement penalties track true model error inconsistently (Lu et al., 2022). Feature-space support distance descends from Mahalanobis out-of-distribution scores (Lee et al., 2018). Berger et al. (2026) report that ensemble disagreement in RSSM rollouts can decrease while decoded physical discrepancy increases, and that prior rollouts can overestimate reward relative to posterior-informed rollouts. This external result motivates separating latent familiarity from trajectory fidelity. It is not matched-intervention evidence for our candidates, and its attractor account is not used as a causal explanation of our PreJEPA results. Our support-relative findings are complementary: internal agreement or proximity to familiar latent regions need not imply accurate physical action effects, and a support-aware ranking score that helps within one family need not transfer to another.

Self-supervised world models. Joint-embedding predictive architectures motivate reconstruction-free latent world models (LeCun, 2022; Assran et al., 2023; Balestriero and LeCun, 2025), and latent dynamics over frozen self-supervised visual features support zero-shot planning (Oquab et al., 2024; Zhou et al., 2025). The LeWorldModel and PreJEPA candidates instantiate this family; the audit treats them as frozen candidates rather than as representation or training proposals.

8 Scope and Limitations

The learned-model evidence covers Cheetah locomotion and Finger Spin, each with one primary scalar query and a five-step physical horizon. An eligibility survey examined 32 systems spanning 13 plausible families; only the five-run PreJEPA family passed every original primary criterion. Released TD-MPC2 sizes and some LeWorldModel checkpoints remain dependent descriptive panels. The evidence therefore supports neither cross-family inference nor a population-level architecture ranking.

A Walker continuation is authorized but not yet terminal, and cannot be pooled with Cheetah or Finger.

The Section 3 results are correlational and single-environment. The size-sweep association uses five dependent released checkpoints from one training recipe; it is a flag, not a selection rule. The anchoring comparison spans two model families with one anchored seed, differing input modalities, and differing native prediction modes; it bounds the claim that task anchoring secures operator fidelity and supports no architecture ranking. Raw scores from the coarse operator diagnostic and the capture-gated audit are not comparable.

Capture is relative to the fixed readout family and does not establish absence or availability under every nonlinear decoder. Probe success also does not show that a planner uses the decoded information (Elazar et al., 2021). Propagation fidelity need not imply high return when planning is discontinuous or action margins are small; the experiments provide no checkpoint-selection result, planning-return improvement, or return guarantee.

The real-model results use literal task queries. The environment-side closure is calibrated synthetically, qualifies only on matched action shells, and its candidate-model decision value remains untested: the first model-side shell audit terminated at the capture gates (Appendix F).

The support-reference analyses remain correlational. Endpoint contraction, motion toward training support, and disagreement–error association do not identify a causal mechanism, and the geometric characterizations of Sections 4.2 and 5 (rotation with excess gain in one family, attenuation in another) constrain candidate mechanisms without selecting one. External evidence that latent rollouts can drift toward familiar regions while physical error grows (Berger et al., 2026) motivates a support-attraction account but was not obtained under matched interventions and is not used as an explanation here. The held-out transfer results bound one frozen additive support rule on one task and bank design; they do not bound every support correction, and the Dreamer studies establish representation boundaries rather than a propagation result, so Dreamer propagation remains uncertified.

9 Conclusion

Planning-time intervention fidelity is a distinct, measurable property of a learned world model. In the settings tested here it is neither revealed by reward fit nor ensured by task-anchored training; it can fail severely while current-state capture, real-effect resolvability, and overall feature alignment all pass; and its severity and geometry are properties of the candidate, task, seed, and support rather than of latent world models as a class. The audit that exposes this treats fidelity as a relation among the model, task query, horizon, intervention support, and native interface: qualified capture–first, read on the surface that actually carries the query, with the model’s own uncertainty trusted only near its training support. Two questions are deliberately left open: which mechanism produces rotation with excess gain in one family and attenuation in another, and whether the audit’s rankings carry decision value for checkpoint selection and return. Both require experiments this framework now makes well-posed.

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A Finite-Horizon Decomposition

The decomposition in Section 2.2 follows by inserting the environment-side closure projection after each model transition, separating the resulting one-step model and closure-approximation residuals, and recursively applying the local stability bound. Its terms distinguish root-capture error, transition error accumulated along the rollout, and target approximation error. No term controls policy discontinuity or a planner’s action margin, so the bound is not a guarantee about realized return.

B Synthetic Calibration Suite

The synthetic suite contains ten cases: exact capture and propagation; exact capture with incorrect dynamics; absent target information; slow persistence; rank deficiency; shuffled actions; stochastic conditional expectations; off-manifold deterministic means; harmless orthogonal basis changes; and forbidden split overlap. The complete suite passes. These cases validate the procedure and stopping rules, not the adequacy of a learned model or the incremental value of empirical closure.

C Per-Candidate Results

Table 2: LeWorldModel confirmatory evaluation. Intervals use complete environment-family bootstrap resamples. The final two columns report the upper endpoint for predicted-effect R^2 and the lower endpoint for predicted-minus-oracle family RMSE.

Checkpoint	Root R^2	Real effect R^2	Predicted effect R^2	R^2 upper	Error diff. lower
s1	0.605	0.127	-5.646	-4.795	0.388
s2	0.609	0.191	-3.364	-2.902	0.289
s3	0.579	0.198	-6.326	-5.349	0.440

Each LeWorldModel checkpoint uses 96 independent evaluation environment families, with three roots and six matched actuator contrasts per family. Checkpoints remain separate descriptive members of one architecture.

Table 3: PreJEPA prospective architecture contrast. Candidate rows remain separate in the source results; the table records only architecture-wide outcomes that hold for all five seeds.

Candidates	Root capture	Real effect R^2	Predicted effect R^2	Oracle deficit
5 seeds	5/5 pass	≈ 0.390	-0.005 to 0.064	5/5 intervals > 0

None of the five PreJEPA candidates passes the strict below-zero distortion condition. The severe LeWorldModel joint failure therefore does not replicate, while accurate propagation is also not established.

Table 4: Finger Spin LeWorldModel evaluation on the complete surface. Intervals use 4,096 whole-family bootstrap resamples on 96 evaluation families. The query is endpoint negative hinge velocity. Raw scores are not comparable to Cheetah torso speed.

Run	Root R^2	Real R^2	Pred. R^2	Pred. 95%	Oracle-err. 95%
01	0.555	0.763	-0.869	$[-1.003, -0.760]$	$[1.368, 1.632]$
02	0.539	0.733	0.287	$[0.212, 0.352]$	$[0.451, 0.637]$
03	0.554	0.789	0.367	$[0.338, 0.391]$	$[0.447, 0.659]$

Run 01 meets the severe below-zero certificate. Runs 02 and 03 retain non-severe positive predicted-effect signal. All three have a strictly positive oracle-relative deficit.

Every held-out LeWorldModel run independently has a strictly negative combined-minus-native interval, so the reciprocal complete-role label is not driven by a single seed.

Table 5: Bidirectional held-out support-augmentation ranking. Spearman ρ uses 384 evaluation families per direction and 4,096 context-stratified whole-family bootstrap draws; the reciprocal direction uses fresh environment-family identities. C is the frozen equal-weight combination of native disagreement percentile and support distance; U is native disagreement; S is support distance alone. Directions are never pooled.

Score	Held-out TD-MPC2 (4M)		Held-out LeWM (epoch 100)	
	ρ	95% interval	ρ	95% interval
Native disagreement U	0.776	[0.762, 0.789]	0.417	[0.381, 0.451]
Support-aware C	0.756	[0.741, 0.770]	0.388	[0.355, 0.420]
Support only S	0.521	[0.498, 0.544]	0.140	[0.102, 0.175]
$C - U$	-0.020	[-0.028, -0.010]	-0.028	[-0.050, -0.007]
$S - U$	-0.255	[-0.278, -0.231]	-0.277	[-0.326, -0.231]

Table 6: Per-run reciprocal-direction ranking on held-out LeWorldModel. Each run is one independently seeded epoch-100 candidate; intervals use the same context-stratified whole-family bootstrap.

Run	ρ_U	ρ_C	$C - U$ (95% interval)
01	0.483	0.436	-0.047 [-0.067, -0.026]
02	0.505	0.453	-0.052 [-0.072, -0.033]
03	0.472	0.431	-0.041 [-0.062, -0.021]

On the shared evaluation, 13 primary combinations meet the inclusion criteria. Candidate and support rows remain separate because every architecture-by-support summary is inconclusive. The support-departure evaluation uses 384 independent families over a 4×4 context-action grid; its two technical replicate runs are paired repeats.

D Probe and Normalization Controls

Target construction, probe training, calibration, and evaluation use disjoint environment-family identities. Target normalization and rank decisions use development data only. Candidate models are unavailable during target construction. Primary real-model tests use family-preserving permutations and 4,096 whole-family bootstrap resamples. Controls include label and feature shuffles, horizon-zero identity, persistence or zero effect, environment-endpoint decoding, wrong action, action null, cyclic actuator, sign swap, and direct feature alignment where defined.

E Dreamer Interface Boundaries

The current-state localization study evaluates four checkpoints, permanently held out of later tests, on 64 fresh validation environment families and 2,048 starting states per checkpoint and representation. Centered posterior logits are the sole surface passing both current-query coordinate gates at every checkpoint. Single sampled categorical states, categorical modes, deterministic recurrent states, and the tested recurrent-state/sample concatenation pass none; posterior probabilities pass only at query-loss scales 16 and 64 (Figure 9). This shows that current information is available after the fact, not that it is propagated.

A separate paired study transfers the centered-logit instrument to two independent `free_nats` pairs and evaluates categorical budgets from one through 1024 samples. Exact-logit instruments pass in all four runs, but no sampled budget passes both coordinates for all runs despite monotone approximation improvement. The intended tight-minus-reference raw-KL direction holds in all training intervals, while both terminal full-prefix contrasts reverse, to +1.406885 and +0.609669; the corresponding one-nat occupancy comparisons are zero. The finite-sample argument and terminal raw-KL direction are therefore not established, and propagation is not tested.

	<i>native surface</i>	checkpoint (query-loss scale)			
		1	4	16	64
posterior logits (centered)		✓	✓	✓	✓
posterior probabilities		✗	✗	✓	✓
categorical sample z		✗	✗	✗	✗
categorical mode		✗	✗	✗	✗
recurrent state h		✗	✗	✗	✗
concat $[h, z_{\text{sample}}]$		✗	✗	✗	✗

Figure 9: **Dreamer current-query surface localization.** Frozen capture-gate outcomes for six native surfaces across the four held-out checkpoints (columns indexed by query-loss scale). Centered posterior logits pass both coordinate gates at every checkpoint; posterior probabilities pass only at scales 16 and 64; the sampled categorical state, its mode, the deterministic recurrent state, and the concatenation pass none. Pass/fail is encoded redundantly by color and glyph. This is current-state availability only; propagation is not tested.

F Empirical-Closure Qualification Boundary

The original Cheetah closure qualification uses 48 development and 48 validation families per support, with supports kept separate. Both supports return no leading component above the significance threshold, and rank zero, so all later target constructions, probes, candidate scores, and evaluation assignments remain unused.

A prospective geometry diagnosis establishes why this rank-zero result cannot be interpreted as absence of action-conditioned structure. The action slots were sampled independently within each family and were therefore exchangeable identifiers rather than common semantic interventions. Action-effect magnitude and the ordering from $H = 1$ through $H = 5$ persist, but common linear predictability is weak and physical dimensionality is not identifiable from the slot SVD. The original rank estimate cannot be used. A valid redesign requires a fresh shared semantic actuator–time dictionary rather than a relaxed rank threshold.

Subsequent environment-side terminals localize that redesigned estimand (E60–E63). The frozen full-quadratic estimator fails over the all-radii action ball and qualifies on a matched fixed-radius shell; a paired-radius test again qualifies both shells but does not confirm task-policy radius conditioning (Figure 10).

The first model-side matched-shell audit applies the qualified shells to three independently trained TD-MPC2 runs with paired 1M/4M snapshots, two supports, and two radii (24 cells). Every primary cell terminates at the capture gates: on task-policy roots, absolute root capture passes for all three 4M runs while paired real-endpoint action-effect capture fails at both radii; on environment-random roots, absolute capture itself fails for all three runs. Each 1M cell receives the same label as its paired 4M cell, and all truth-fidelity, oracle-relative, and imagined-geometry labels are uncertified by capture failure. A readout qualified for broad 17-dimensional state reconstruction on a disjoint development bank guaranteed neither support transfer for the scalar query nor sufficient resolution after endpoint-minus-root differencing; absolute-query capture and paired-effect resolution are distinct instrument requirements.

G Inferential Breadth

The eligibility survey examines 32 systems spanning 13 plausible architecture families. Only the five-run PreJEPA family satisfies every original primary criterion; four runs are primary and one is an extra run. Five released TD-MPC2 sizes remain descriptive examples rather than independent

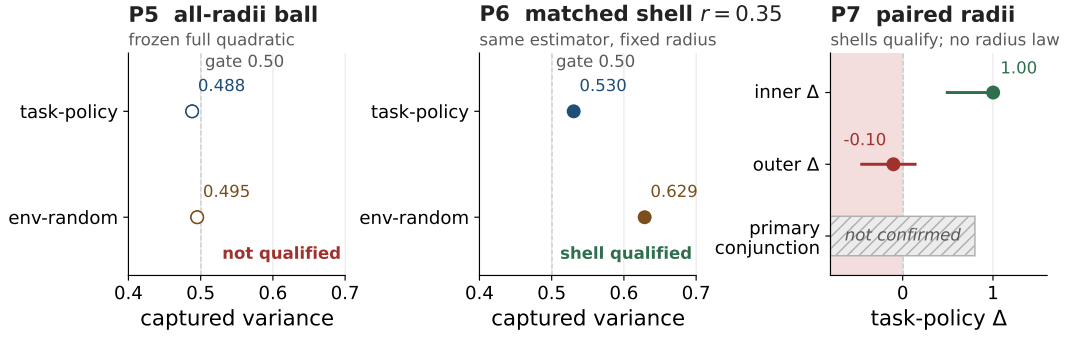


Figure 10: **Environment-side shell localization.** Successive C2 terminals on Cheetah (E60–E63). The frozen full-quadratic estimator fails the all-radii action-ball gate on both supports (P5) and qualifies on the matched $r = 0.35$ shell on both supports (P6). A paired inner/outer shell test again qualifies both shells (P7), but the primary task-policy radius-conditioning contrast is not confirmed. Supports are analyzed separately; environment-random cannot rescue the primary arm. This is environment-side estimand localization only: it does not establish candidate-model capture, propagation, decision validity, regret, or return.

training runs. The survey therefore supports no cross-family inference, monotonic size law, or model ranking.

The separate exploratory TD-MPC2 panel contains 80 complete scores. The 317M model combines accurate capture with extreme predicted action-effect error. A 19M model on policy support outperforms persistence, while the same candidate is worse under environment-random starting states. These contrasts establish support sensitivity within a dependent released-size panel, not confirmatory architecture evidence.